5D_b has a curled circle, S^1.

Part a has an off-diagonal element $g_{04}=\epsilon c p_4 h(r)$.

Part b has an off-diagonal element $g_{04}=-\frac{c\, p_4\, m \sinh 2\,\beta}{r}$.

Notation and Setup:

Starting Metric for this Case: $g_{00}=c^2\left(1-\frac{2m}{r}\right)$, $g_{04}=c\,\epsilon\,h[r]\,p_4$, $g_{11}=-\frac{1}{1-\frac{2m}{r}}$, $g_{22}=-r^2$, $g_{33}=-r^2\,\text{Sin}[\theta]^2$, $g_{40}=c\,\epsilon\,h[r]\,p_4$, $g_{44}=-p_4^2$

Corresponding Line Element

$$ds^2 = c^2\left(1-\frac{2m}{r}\right)dt^2 + c\,\epsilon\,h[r]\,p_4\,dt\,d\psi_4 - \frac{1}{1-\frac{2m}{r}}dr^2 - r^2\,d\theta^2 - r^2\,\text{Sin}[\theta]^2\,d\phi^2 + c\,\epsilon\,h[r]\,p_4\,d\psi_4\,dt - p_4^2\,d\psi_4^2$$

Output Display Simplification - Parameters are sometimes suppressed
Prime (') means derivative wrt r.
Dot (.) means derivative wrt t.
Examples: $A = A(t,r)$, $A' = \partial_r A(t,r)$ and $\dot{A} = \partial_t A(t,r)$

Lambda vacuum equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$.

They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$.

a: Coulomb's Law

The line element above shows the initial assumptions we have made:

there is no scale factor a(t),

$A(r) = c^2\left(1-\frac{2m}{r}\right)$,

$B(r) = \frac{c^2}{A[r]}$, as in the Schwarzschild equations,

$C(r) = 1$,

$D_4(r) = 1$,

and off-diagonal metric components

$g_{04} = g_{40} = c\,\epsilon\,h[r]\,p_4$ where $\epsilon \approx 0$ and $h(r)$ is a free parameter.

We also assume

$\Lambda = 0$.

Using this starting point, the non-zero Lambda field equations are

$$R_{00}: \quad \frac{h^2 m^2 \epsilon^2}{(2m-r)\, r^2 \left(-2m+r+h^2 r \epsilon^2\right)} - \frac{h m \epsilon^2 h'}{(2m-r)\left(-2m+r+h^2 r \epsilon^2\right)} + \frac{2 h m^2 \epsilon^2 h'}{(2m-r)\, r \left(-2m+r+h^2 r \epsilon^2\right)} + \frac{2 m^2 \epsilon^2 (h')^2}{(2m-r)\left(-2m+r+h^2 r \epsilon^2\right)} - \frac{2 m r \epsilon^2 (h')^2}{(2m-r)\left(-2m+r+h^2 r \epsilon^2\right)} + \frac{r^2 \epsilon^2 (h')^2}{2(2m-r)\left(-2m+r+h^2 r \epsilon^2\right)} = 0$$

$$R_{04}: \quad -\frac{h'}{h\left(-2m+r+h^2 r \epsilon^2\right)} - \frac{4 m^2 h'}{h r^2 \left(-2m+r+h^2 r \epsilon^2\right)} + \frac{4 m h'}{h r \left(-2m+r+h^2 r \epsilon^2\right)} - \frac{h \epsilon^2 h'}{-2m+r+h^2 r \epsilon^2} + \frac{3 h m \epsilon^2 h'}{2 r \left(-2m+r+h^2 r \epsilon^2\right)} + \frac{2 m h''}{h\left(-2m+r+h^2 r \epsilon^2\right)} - \frac{2 m^2 h''}{h r \left(-2m+r+h^2 r \epsilon^2\right)} - \frac{r h''}{2 h \left(-2m+r+h^2 r \epsilon^2\right)} + \frac{h m \epsilon^2 h''}{-2m+r+h^2 r \epsilon^2} - \frac{h r \epsilon^2 h''}{2\left(-2m+r+h^2 r \epsilon^2\right)} = 0$$

$$R_{11}: \quad \frac{3 h^2 m^2 \epsilon^2}{r^2 \left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{2 h^2 m \epsilon^2}{r\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{2 h^4 m \epsilon^4}{r\left(-2m+r+h^2 r \epsilon^2\right)^2} + \frac{h m \epsilon^2 h'}{\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{2 h m^2 \epsilon^2 h'}{r\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{h^3 m \epsilon^4 h'}{\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{2 m^2 \epsilon^2 (h')^2}{\left(-2m+r+h^2 r \epsilon^2\right)^2} + \frac{2 m r \epsilon^2 (h')^2}{\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{r^2 \epsilon^2 (h')^2}{2\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{h^2 m r \epsilon^4 (h')^2}{\left(-2m+r+h^2 r \epsilon^2\right)^2} + \frac{h^2 r^2 \epsilon^4 (h')^2}{2\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{4 h m^2 \epsilon^2 h''}{\left(-2m+r+h^2 r \epsilon^2\right)^2} + \frac{4 h m r \epsilon^2 h''}{\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{h r^2 \epsilon^2 h''}{\left(-2m+r+h^2 r \epsilon^2\right)^2} + \frac{2 h^3 m r \epsilon^4 h''}{\left(-2m+r+h^2 r \epsilon^2\right)^2} - \frac{h^3 r^2 \epsilon^4 h''}{\left(-2m+r+h^2 r \epsilon^2\right)^2} = 0$$

$$R_{22}: \quad \frac{h^2 m \epsilon^2}{r^2 \left(-2m+r+h^2 r \epsilon^2\right)} - \frac{h \epsilon^2 h'}{-2m+r+h^2 r \epsilon^2} + \frac{2 h m \epsilon^2 h'}{r\left(-2m+r+h^2 r \epsilon^2\right)} = 0$$

$$R_{33}: \quad \frac{h^2 m \epsilon^2}{r^2 \left(-2m+r+h^2 r \epsilon^2\right)} - \frac{h \epsilon^2 h'}{-2m+r+h^2 r \epsilon^2} + \frac{2 h m \epsilon^2 h'}{r\left(-2m+r+h^2 r \epsilon^2\right)} = 0$$

$$R_{44}: \quad -\frac{2 m \epsilon^2 (h')^2}{4m-2r-2h^2 r \epsilon^2} + \frac{r \epsilon^2 (h')^2}{4m-2r-2h^2 r \epsilon^2} = 0$$

R_{33} is redundant, the same as R_{22}.

Keeping only first-order terms [i.e., dropping terms of order ϵ],

R_{00}, R_{11}, R_{22}, and R_{44} become 0=0 identities, leaving only R_{04}:

$R_{00}: \quad 0 = 0$

$$R_{04}: \quad -\frac{h'}{h(-2m+r)} - \frac{4 m^2 h'}{h r^2 (-2m+r)} + \frac{4 m h'}{h r (-2m+r)} + \frac{2 m h''}{h(-2m+r)} - \frac{2 m^2 h''}{h r (-2m+r)} - \frac{r h''}{2 h (-2m+r)} = 0$$

$R_{11}: \quad 0 = 0$

$R_{22}: \quad 0 = 0$

$R_{44}: \quad 0 = 0$

R_{04} can be simplified by finding a common denominator and setting the numerator to zero:

$$R_{04}: \quad -2\,(-2\,m+r)^2\,h' - (2\,m-r)^2\,r\,h'' = 0$$

Factoring out the polynomial $(2m-r)^2$, this becomes

$$R_{04}: \quad -h' - \frac{r\,h''}{2} = 0$$

$$\Rightarrow \qquad h(r) = C_1 + \frac{C_2}{r} \quad \text{where } C_1 \text{ and } C_2 \text{ are constants of integration.}$$

- **C_2/r is the Coulomb potential**, the electric potential of a point charge, with C_2 proportional to the charge.
  - ✓ We have derived Coulomb's law from Einstein's vacuum equations in five dimensions.

- **C_1 is a change of coordinates**, $\psi \to \psi + \lambda t$.
  - ✓ In electromagnetism, adding a constant to the potential changes nothing physical.
  - ✓ It is called a gauge choice.
  - ✓ In KK theory, that **gauge freedom is literally a coordinate freedom along the circle.**

b: Verify a generalized solution

The Part b line element is

$$ds^2 = c^2\left(1 - \frac{2\,m\,\text{Cosh}[\beta]^2}{r}\right)dt^2 - \frac{c\,m\,\text{Sinh}[2\beta]\,p_4}{r}\,dt\,d\psi_4 - \frac{1}{1-\frac{2m}{r}}\,dr^2 - r^2\,d\theta^2$$
$$- r^2\,\text{Sin}[\theta]^2\,d\phi^2 - \frac{c\,m\,\text{Sinh}[2\beta]\,p_4}{r}\,d\psi_4\,dt - \left(1 + \frac{2\,m\,\text{Sinh}[\beta]^2}{r}\right)p_4^2\,d\psi_4^2$$

The line element above shows the initial assumptions we have made:

there is no scale factor $a(t)$,

$$A(r) = c^2\left(1 - \frac{2\,m\,\text{Cosh}[\beta]^2}{r}\right),$$

$$B(r) = \frac{1}{1-\frac{2m}{r}} \quad \text{as in the Schwarzschild equations,}$$

$$C(r) = 1,$$

$$D_4(r) = 1 + \frac{2\,m\,\text{Sinh}[\beta]^2}{r},$$

and off-diagonal metric components

$$g_{04} = g_{40} = -\frac{c\,m\,\text{Sinh}[2\beta]\,p_4}{r}.$$

We also assume

$$\Lambda = 0.$$

Connection to Part a.

For small β, sinh $2\beta \approx 2\beta$, so $g_{04} \approx -2c\, p_4\, m\beta/r$. That is Part a's $\varepsilon\, c\, p_4\, h(r)$ with $h = C_2/r$ and $\varepsilon C_2 = -2m\beta$. The linear Coulomb solution of Part a is the small-β limit of this exact solution.

Not the ordinary Reissner-Nordström black hole.

D_4 now depends on r, so the circle's size varies with r. In 4D language, that varying size is a scalar field (the dilaton). The result is a charged black hole with a scalar field, not the ordinary 4D charged black hole.

With this metric, Λ=0 field equations are all satisfied:

R_{00}: 0 = 0

R_{01}: 0 = 0

R_{02}: 0 = 0

R_{03}: 0 = 0

R_{04}: 0 = 0

R_{11}: 0 = 0

R_{12}: 0 = 0

R_{13}: 0 = 0

R_{14}: 0 = 0

R_{22}: 0 = 0

R_{23}: 0 = 0

R_{24}: 0 = 0

R_{33}: 0 = 0

R_{34}: 0 = 0

R_{44}: 0 = 0

Notes:

β = 0: the uncharged black string:

With $\beta = 0$, $A = c^2(1 - 2m/r)$, $D_4 = 1$ and $g_{04} = 0$. That is the Schwarzschild metric times a fixed circle, the same background used in Part a. It is called a black string because the horizon at r = 2m extends around the whole circle.

Why it is Ricci-flat for every β:

Write

$T = ct$ and $Y = p_4\psi$.

The uncharged string is then

$ds^2 = dT^2 - dY^2 - (2m/r)\, dT^2 - (\text{spatial part})$.

Now apply a Lorentz boost[1]:

$$T = T' \cosh\beta + Y' \sinh\beta$$
$$Y = T' \sinh\beta + Y' \cosh\beta$$

The term $dT^2 - dY^2$ doesn't change under the boost. The term $-(2m/r)dT^2$ becomes $-(2m/r)(\cosh\beta\, dT' + \sinh\beta\, dY')^2$. Expanding it gives exactly the Part b metric:

$$g_{00} = c^2\left(1 - \frac{2m\cosh^2\beta}{r}\right)$$
$$g_{44} = -p_4{}^2\left(1 + \frac{2m\sinh^2\beta}{r}\right)$$
$$g_{04} = -\frac{c\,p_4\,m\sinh 2\beta}{r}$$

A coordinate change can't create curvature. So $R_{\mu\nu} = 0$ for every β, which is what the notebook confirms.

β is the rapidity[1]:

The boost velocity along the circle is $v = c\tanh\beta$. Cosh and sinh appear because a boost keeps T^2-Y^2 fixed, just as a rotation keeps x^2+y^2 fixed using cos and sin.

This boost[1] is not simply a coordinate change:

Because ψ is periodic, ψ and $\psi+2\pi$ are the same point, but the boost gives them two different times t'. So the boost is a coordinate change only locally; globally, the boosted string is a different spacetime.

The boosted string has angular momentum because, relative to the original frame, it moves around the circle.

(Footnote)

1 **Boost and Rapidity** - A rotation in 3-space is a coordinate change where the rotated coordinates are represented in terms of an angular change, say θ.

Space and time coordinates cannot be rotated into each other the way space axes can. Mixing them gives a boost (cosh, sinh) instead of a rotation (cos, sin).

Like a rotation, a boost can be a change of coordinates, but it is described by velocity instead of an angle. The size of a boost, say β, is called a rapidity. It is a "hyperbolic angle". The corresponding speed is $v=c\tanh\beta$.

Boost represents electric charge:

In Kaluza-Klein theory, momentum around the circle is electric charge. So boosting the uncharged string makes it charged, with charge proportional to m sinh 2β.